RDIMS# 1184210

ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Naval Electronic Sensor Operator (00115)
FC Operator/ CIWS Operator

RQ-NESOP-S2

NQUAL - ALAY

MITE ID
122604

RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to PO standard are to be requested through NTDC(A) and approved by CNPTG. Ammendments to Lesson Guidence are to be approved by CO NTDC(A).

Revision	Approval Date	Authority
2.9	07 Jun 2021	NPTG HQ

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FOREWORD

1. This training is based on the specification found at :

   a. A-PD-055/PP-002 Job Based Specification Naval Electronic Sensor Operator dated: 6 May 2019.

2. A trainee will achieve the qualification **ALAY** when the standards listed in all Performance Objectives have been met.

3. A trainee will achieve the MITE Course IDs 122604 when standards listed in the associated Performance Objectives have been met:

4. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A) / NFS (P)	All	122604	ALAY

CHAPTER 1

TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the NESOp Occupation for the S2 Rank Qualification ALAY. This QSP is designed to assist NTDC(A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, Annex B. For this course, a period of instruction is 50 minutes in duration. The recommended schedule is based on **maximum load**. Scheduling cells are to increase/decrease course length based on number of students enrolled on the course.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	170
Enabling Checks (ECs)	27
Performance Checks (PCs)	18
Actual Training Time (ATT) Sub-total	27 **trg days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Annex B Table 3.1	
Base Administrative Time (BAT)	2
Training Administrative Time (TAT)	2
Training Travel Time (TTT)	5
Physical Fitness (PT)	18
Sub-total	4 **trg days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	31 **trg days**

REQUIRED COURSE STAFFING

3. Idenfity key personnel required to conduct the training.

Position	Rank	Qualifications	Qty	PO / EO
Course Director	LT(N)	ALBU	1	All
Instructor	MS	ALAZ	1 at optimum load 2 if more than 8 students	All

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 12;

   b. Minimum – 6; and

   c. Optimum – 8.

PREREQUISITES

5. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

   a. RQ-OS (ALFS); and

   b. Have completed one year post QL3 employed.

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COURSE REPORTING

6. Upon completion of the course, students shall receive a paper record of completion by means of form CF 377 Course Report and Form CF 289 (or locally produced equivalent) Certificate of Military Achievement IAW DAOD 31-9 and NPTORD 4500-51.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

7. PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

8. Nil.

CHAPTER 2

PERFORMANCE OBJECTIVES & LESSON GUIDANCE

NPTG HQ Approval – 07/06/2021
Version No: 2.9
MITE Crse ID : 122604

PO 001 – Operate Combat Systems Equipment: CEROS

1. **Performance**: Operate Combat Systems Equipment: CEROS (B00080, D00060, D00010, D00080)

2. **Conditions**:

   a. Given:

      (1) Resources;
      CEROS worksheet
      CPF layout plan
      Screenshot of faults (system overview)

      (2) References;
      NCPM 1000
      NAVORD 4995-0
      SEMS
      57mm mk 3 Hand Book
      CEROS 200 manual
      (3) Supervision.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) At Sea/Alongside Day or Night
      (2) Trainer/Simulator Day or Night

3. **Standard**: The Fire Control Operator shall operate combat systems equipment: CEROS (B00080, D00060, D00010, D00060) by:

   a. Executing preparation for live firing drills and exercises by:

      (1) Conducting closing up, clearing away procedures IAW NCPM 1000;
      (2) Conducting verbal misfire drills;
      (3) Assuming policy to include:
         (a) Adhering to EMCON restrictions (B00120); and
         (b) Setting parameters set by the SWC;
      (4) Posting weapons firing out-of-bounds signs; and
      (5) Preparing surface targets for gunnery exercises (D00020);

   b. Executing gunnery drills and misfire procedures for 57mm by:

      (1) Conducting 57MM Pre-Action Calibration;

(2) Operating CEROS for gunnery drills by:
- (a) Conducting system tracking assessments to include:
  - i. Carrying out EP measures (C00070) by:
    - (1) Assessing jamming (C00050); and
    - (2) Reacting to changes in the EW environment (C00580)
- (b) Communicating reports;
- (c) Performing threat analysis (C00540);
- (d) Identifying faults to include:
  - i. Check load;
  - ii. Misfire; and
  - iii. General faults;
- (e) Engage in live firing; and
- (f) Conducting naval fire support drills.

c. Execute missile drills and misfire procedures by:

(1) Operating CEROS for missile drills by:
- (a) Conducting system tracking assessments to include:
  - i. Carrying out EP measures (C00070) by:
    - (1) Assessing jamming (C00050); and
    - (2) Reacting to changes in the EW environment (C00580)
- (b) Communicating reports;
- (c) Performing threat analysis (C00540);
- (d) Identifying faults to include:
  - i. Misfire; and
  - ii. General faults.
- (e) Engage in live firing

4. **Training Limitations**: Nil.

EO 001.01 - Execute preparation for 57mm gunnery drills

001.01.01 – Intro to CEROS

Conditions		Sequence
Unclassified classroom		N/A
Time	**Classification**	**Instructor**
200 min	Unclassified	RQ-S2 Instructor

Learning Objectives	Method
1) What is CEROS 2) How do we employ CEROS 3) CEROS system caps/lims 4) CEROS vulnerabilities 5) Weapons safety	Think Pair Share Lecture Team Quiz

Activity Steps

1) Instructor to distribute a CEROS worksheet to each student. Explain to students that they are to fill out their worksheet throughout the entire Performance Objective lessons. These will be used throughout the course.
2) Instructor to introduce lesson for CEROS, explaining what the system is for "the system used to guide the ESSM and control the 57mm gun". May expand on what is CEROS and allow students to provide some of their own experience. (5 min)
3) Instructor ask students to individually think about and write down their answer to the following question "based on your experience, how do we employ CEROS?" Students are provided 5 min to create their own list (5 min)
4) Students are then asked to pair up and share their lists between them. Students are to discuss the differences between their two lists. (5 min)
5) Student pairs are to then share their responses with the class, and instructor to create one class list on the white board. Instructor to provide input to ensure the list is complete. (10 min)
6) Instructor to conduct a PowerPoint lesson explaining the capabilities, limitations and vulnerabilities of CEROS. (100 min)
    - Min/max frequencies
    - Train & elevation rates
    - Environmental restrictions
    - Ranges
    - TV camera/IR camera/laser
    - EPM features
    - CWI
    - RADAR modes
    - Methods of tracking (optronic/IR, active RADAR)
    - Vulnerabilities (FCER 1&2, CMS, operator, blind spots, RADAR conditioning, weather)

	7) Instructor will then ask each student to create 3 questions (and the responses) based on the content that was just presented. Once complete, they are to provide these questions/answers to the instructor. (5 min) 8) Students are then broken into two teams and the instructor is to use the questions created by the students as a quiz to the class for competition points (20 min)

Resources	References
1) CEROS worksheet 2) PowerPoint lesson 3) DWAN computer 4) Projector 5) Weapons Safety Powerpoint	1) CEROS operators manual

Notes to Instructor

EO 001.01 - Execute preparation for live firing drills and exercises

001.01.02 – Levels of Readiness and States of Preparedness

Conditions		Sequence
Unclassified classroom		Following EO 001.01.01
Time	**Classification**	**Instructor**
50 min	unclassified	RQ-S2 Instructor

Learning Objectives	Method
1) Degrees of readiness 2) States of preparedness	Student readings Group presentations

Activity Steps	
	1) Students are provided paper print outs of the readings (SSOs paras on States of Preaparedness (6.10.2 version 10.1) and Degrees of readiness 1st-5th (6.10.1 version 10.1). Students are provided 20 min to conduct readings (20 min) 2) Upon completion of the readings, class is to be split into 4 groups. Groups will be assigned a subject where they must prepare a flip chart to present their information, expanding on the application of their topic and its importance. (10 min) Topics • 1st Degree of Readiness • 2nd Degree of Readiness • 3rd Degree of Readiness • States of Preparedness 3) Groups to present their topic to the class with instructor expanding on information as necessary. (20 min)

Resources	References
1) SSOs readings pages 612-618 handout 2) Flip Charts (x4) and markers	1) SSOs 2) ATP 1 vol 1

Notes to Instructor

EO 001.01 - Execute preparation for live firing drills and exercises

001.01.03 – EMCON and RADHAZ

Conditions		Sequence
Unclassified classroom		Following EO 001.01.02
Time	**Classification**	**Instructor**
50 min	Unclassified	RQ-S2 Instructor

Learning Objectives	Method
1) Understand EMCON restrictions for Operations for CEROS 2) Actions taken to assume RADHAZ precautions for CEROS	Case Study Student Reading Interactive Lecture

Activity Steps	
	1) Students are provided a case study to read on an EMCON/RADHAZ scenario. (5 min) 2) Instructor to lead a class discussion on the scenario, having students identify: • "what went wrong" • "what should have occurred" • "what are the implications of these failings" (10 min) 3) Students will then read SEMS Annex B to S9 Appendix B4 (positive control regarding FWD/AFT CEROS and CIWS) (15 min) 4) Instructor to review HERP/HERO/HERF and how it applies to CEROS operations. They can provide examples of when RADHAZ precautions go into effect and how to assume RADHAZ safe for CEROS. Describe positive control of the RC equipment. (15 min)

Resources	References
1) Case Study reading 2) SEMS reading package	1) SEMS **2)** NAVORD 4995-0

Notes to Instructor

EO 001.01 - Execute preparation for live firing drills and exercises

001.01.04 – Out of Bounds Signage

Conditions		Sequence
Unclassified classroom		Following EO 001.01.03
Time	**Classification**	**Instructor**
50 min	Unclassified	RQ-S2 Instructor

Learning Objectives	Method
1) Safety precautions for 57mm and ESSM firing 2) Requirements for posting OOB signage 3) Safety Pipes for weapons firing	Student reading Confirmation questions Diagram Labeling

Activity Steps	
	1) Students to read NAVORD 4995-0 section 3 (Internal safety) (25 min) 2) Instructor to conduct a reading summary with the class with follow up confirmation questions (10 min) • Why do we put areas OOB during weapons firing? • Who is responsible for AWW weapon safety? • What role do you play as an RCO during weapons firing? • Who is responsible to ensure that the OOB areas are piped? 3) Students are then provided a print out of a CPF plan. Students are to indicate where OOB signs for various weapons firing serials go on a plan of a CPF and create a list. (15 min) ***** reminder to students to study the OOB signage locations for EC 001.01**

Resources	References
1) NAVORD 4995-0 section 3 readings 2) CPF print plan	1) NAVORD 4995-0 2) NCPM 1000

Notes to Instructor

EO 001.01 - Execute preparation for live firing drills and exercises

001.01.05– Prepare Surface Targets

Conditions		Sequence
Trainer/Ship		Following EO 001.01.04
Time	**Classification**	**Instructor**
200 min	Unclassified	Target Cell Staff

Learning Objectives	Method
1) Types of Targets used by RCN a. Towed air targets b. Towed surface targets c. Remote controlled targets d. Free floating targets 2) Setting up surface targets 3) Safety considerations for target preparation	Lecture Demonstration Practical

Activity Steps	
	1) Students to travel to target cell for a facility famil. (25 min) 2) Target cell staff to explain the types of targets used by the RCN.(50 min) 3) Students are to setup and teardown an HSITT (or other target) with instructor/target cell guidance. Instructors to provide guidance on safety considerations when preparing surface targets. (100 min) 4) Students to travel back to training establishment. (25 min)

Resources	References
1) Transport to and from Target Cell 2) HSITT	1) Killer red Tomato PDF 2) HSITT PDF 3) Hammerhead PDF

Notes to Instructor

EO 001.01 - Execute preparation for live firing drills and exercises

001.01.06 – CEROS System Manipulation

Conditions		Sequence
SEACOT		Following EO 001.01.05
Time	**Classification**	**Instructor**
150 min	SECRET	RQ-S2 Instructor

Learning Objectives	Method
1) CEROS buttonology 2) CEROS system setup	EDI Practical

Activity Steps

1) Instructor will explain the basic intro to CEROS system setup requirements to the class to include:
   - FCD control
   - FCD setting
   - Gun engagement
   - Gun control
   - ESSM control
   - ESSM engagement
   - System control
   - Console

   Explain what is contained in each page, when you are required to use each page and how to manipulate through the pages. (5 min)

2) Instructor will demonstrate a walk through of the buttonology and system setup of CEROS. (80 min)
3) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (40 min)
4) Students will then conduct a CEROS system setup independently with instructor support (explore the system). (25 min)

Resources	References
	1) CEROS user manual

Notes to Instructor

EO 001.01 - Execute preparation for live firing drills and exercises

001.01.07 – CUCA and Assuming Policy

Conditions		Sequence
SEACOT		Following EO 001.01.06
Time	**Classification**	**Instructor**
200 min	SECRET	RQ-S2 Instructor

Learning Objectives	Method
1) Conducting Closing Up and Clearing Away 2) Schedule of Verbal misfire drills 3) Assuming Policy a. Setting SWC parameters b. System manipulation c. Inputting Policy orders (SALVO, Red Alert, etc)	EDI Practical

Activity Steps	
	1) Instructor will explain CUCA and Assuming Policy to the class. They will also explain that verbal misfire drills are normally done prior to assuming Policy as a cultural norm but this will not be done in this lesson as it is not doctrine and students will learn drills in future lessons. (5 min) 2) Instructor will demonstrate a walk through of conducting CUCA and Assuming Policy. (15 min) 3) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (40 min) 1) Students will then conduct CUCA and assumption of Policy independently with instructor support and use of NCPM 1000. (140 min)

Resources	References
1) NCPM 1000	1) NCPM 1000

Notes to Instructor

EC 001.01 - Execute preparation for 57mm gunnery drills

Scope	Conditions	Sequence
EO 001.01	SEACOT	Following EO 001.01
Time	**Classification**	**Instructor**
250 min total	Unclassified	RQ-S2 Instructor

Assessment Steps	Input: Students will be provided a CEROS console and support manning detailed in NWTP 2, where they must execute the preparation duties of FCO for surface engagements. Performance: students will conduct a 15 min run, where each run will include: • CUCA • Assume policy • Verbally indicate OOB signage locations for both 57mm and ESSM firing Output: The trainee must achieve 10/12 SATs to pass this EC. Feedback: Immediate verbal and written feedback provided using EC 001.01 checklist. 5 min allocated to reset the scenario and conduct debrief.

Resources	References
1) NCPM 1000 2)	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor

Limitations	Unable to physically post the signage, only able to assess knowledge of locations verbally
Impact of Failure	Retrain and reassess

Item	EC 001.01 EXECUTE PREPARATION FOR 57MM GUNNERY DRILLS	SAT	UNSAT	Remarks
1. Conduct Close Up Cleared Away IAW NCPM 1000				
Conduct Communications Check		☐	☐	
Conduct system setup IAW NCPM 1000		☐	☐	
Input environmentals and Nav data		☐	☐	
System Display setup IAW NCPM 1000		☐	☐	
Assign gun to FC channel		☐	☐	
Reports Closed up Cleared away to FC Sup		☐	☐	
2. Input Policy data to include:				
Select magazine		☐	☐	
Select forced recoil		☐	☐	
Set salvo size		☐	☐	
Assign missile		☐	☐	
3. Verbally indicate OOB signage locations for weapons firing				
Verbally indicate **ALL** 57 mm signage locations ☐ Fwd FX hatch ☐ Aft FX hatch ☐ Fwd port breezeway door ☐ Bridge Wing Doors x 2		☐	☐	
Verbally indicate **minimum 10** ESSM signage locations ☐ Fwd FX hatch ☐ Aft FX hatch ☐ Fwd port breezeway door ☐ Bridge Wing Doors x 2 ☐ Aft Port breezeway door ☐ sea head flat doors x 2 ☐ port breezeway hatch from 1 to 01 deck ☐ port missile deck door ☐ fwd hangar doors x 2 ☐ LSO CMPT hatch ☐ Hangar lobby doors x 2 ☐ AX hatch		☐	☐	
Score: /12	The trainee must achieve 10/12 SATs to pass this EC.			

Positive Feedback		**Improvement Feedback**	
Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 001.02 - Execute gunnery drills and misfire procedures for 57mm

001.02.01–57mm Pre-Action Calibration

Conditions		Sequence
Unclassified classroom		Following EC 001.01
Time	**Classification**	**Instructor**
100 min	unclass	RQ-S2 Instructor

Learning Objectives	Method
1) Purpose of PAC 2) Types of PAC 3) Principles of spotting and corrections 4) Basics of ballistics and trajectory (Coriolis effect, gravity, environmental conditions) 5) How to calculate corrections	Essence Game Demonstration Lesson Practical

Activity Steps	
	1) Instructor to ask students to pair up and between the two of them create and write down a definition for “What is the purpose of a PAC” in less than 32 words. (5 min) 2) Students are then asked to cut/alter their definition to 16 words or less. (2 min) 3) Finally students are asked to reduce their definition to 8 words or less. (3 min) 4) Instructor will then explain the different types of PACs (deliberate, urgent, and non MVR deliberate). (5 min) 5) Instructor will explain the purpose of conducting spotting and corrections. They will refer to website in instructor resources (LSC to develop similar training scenario) (intro) and have students direct how to move the cannon and change the speed in order to hit the target. (20 min) 6) Instructor to then conduct a demonstration lesson on how to calculate corrections using the whiteboard. They must also explain the PAC elevation correction table (when to use and how to use). (20 min) 7) Instructor will then do more examples getting students to assist in conducting the calculations for error of the day. (15 min) 8) Instructor will hand out a worksheet where students are to use the provided data to determine the error of the day. When everyone is done, have them check their answers with a peer. Then instructor to provide DS solution and explain further if students encountered difficulties. (30 min)

Resources	References
1) DWAN workstation 2) Projector 3) 57mm range corrections table (NCPM 1000) 4) PAC elevation correction table 5) White board & markers 6) Corrections worksheet 7) Corrections worksheet DS solution 8) Refer to website in instructor resources	1) CFCD 106 MARTI 3221 2) NCPM 1000

EO 001.02 - Execute gunnery drills and misfire procedures for 57mm

001.02.02 –57mm Caps/Lims and Ammunition Types

Conditions		Sequence
Unclassified classroom		Following EO 001.02.01
Time	**Classification**	**Instructor**
50 min	unclassified	RQ-S2 Instructor

Learning Objectives	Method
1) 57 mm gun capabilities and limitations 2) 57 mm functionality 3) 57 mm ammunition types	Lecture Card sort

Activity Steps

1) Instructor to present a PowerPoint lecture on the caps/lims and functionality of 57mm gun. They will explain the different types of ammunition in this lecture. (35 min)

2) Students are then broken into teams of 2-3 students. Each group is given a set of cards labeled:
   a. PFHE
   b. HCER
   c. BL/P
   d. NFVT
   e. Cycling/Dummy
   f. 3P

   Students are to tape the cards up on the wall in an order of their choice. (5 min)
3) Groups will then be provided a second set of cards with pictures of the types of rounds. They must match the card to the appropriate name card, tapping it under the ammunition type. (Instructor to provide feedback prior to moving to next step). (5 min)
4) Groups are provided a third set of cards on the <u>use</u> of the ammunition type which must be matched to the ammunition type/picture already taped to the wall. (5 min)
   Cards will be labeled:
   a. Air Round
   b. Surface Round
   c. Training
   d. Air Training
   e. Maintenance, Feed system test

5) Instructor to check the groups card sort to confirm correctness

Round	Colour	Lettering/marking	Ballistics Code
PFHE	olive drab	yellow	AM 1
HCER	olive drab	yellow	AM 2
BL/P	light blue	white	AM 1
NFVT	light blue	white lettering with brown band or white lettering with yellow band	AM 1
Cycling	bronze	none or DUMMY in black letters	N/A

Table 4 Ammunition Table

Resources

1) DWAN computer
2) Projector
3) PowerPoint lecture (57mm)
4) Card Sort Cards (x3) & tape

References

1) NAVORD 4995-0
2) 57mm Mk 3 Handbook

Notes to Instructor

EO 001.02 - Execute gunnery drills and misfire procedures for 57mm

001.02.03 –57mm Famil

Conditions		Sequence
Trainer/Ship		Following EO 001.02.02
Time	**Classification**	**Instructor**
150 min	unclassified	RQ-S2 Instructor

Learning Objectives	Method
1) 57 mm functionality	Facility Visit

Activity Steps	
	1) Students to travel to Trainer/Ship(25 min) 2) Trainer/Ship staff to conduct a lesson to demonstrate the 57mm functionality, loading, parts/components, feed system, cassettes, emergency stops and mechanically safe switch. (100 min) 3) Student to travel back to base (25 min)

Resources	References
1) transport to/from Trainer/ship	1) 57 mm mk3 handbook

Notes to Instructor

EO 001.02 - Execute gunnery drills and misfire procedures for 57mm

001.02.04 –57mm PAC

Conditions		**Sequence**
SEACOT		Following EO 001.02.03
Time	**Classification**	**Instructor**
700 min	SECRET	RQ-S2 Instructor

Learning Objectives	**Method**
1) PAC buttonology 2) PAC screen layout/configuration 3) PAC drills (deliberate, urgent, and non MVR deliberate)	EDI Practical

Activity Steps	
	1) Instructor will explain 57mm PAC to include: • Buttonology • Screen layouts • PAC drills Explain what is contained in each page, when you are required to use each page and how to manipulate through the pages. (10 min) 2) Instructor will demonstrate a walk through of the buttonology and system setup, then demo all PAC drills. (30 min) 3) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (30 min) 4) Students will then pair up and conduct PAC drills with partners using NCPM 1000 with instructor support. Instructor to circulate and watch each student conduct drills and provide feedback. (630 min)

Resources	**References**
1) NCPM 1000	1) NCPM 1000

Notes to Instructor

EO 001.02 - Execute gunnery drills and misfire procedures for 57mm

001.02.05 – Investigate/Aquire

Conditions		**Sequence**
SEACOT & SEACOT briefing room		Following EO 001.02.04
Time	**Classification**	**Instructor**
350 min	SECRET	RQ-S2 Instructor

Learning Objectives	**Method**
1) conduct contact investigate 2) perform threat analysis (report a contact) 3) means to range a contact a. laser range finder b. RADAR 4) means of acquiring contact a. Passive RADAR b. Auto c. Optronic d. Manual Optronic e. Passive RADAR f. Active RADAR With Jammer 5) Assess tracking (locked- on, A-Scope, FCD status/smooth tracking) 6) Carry out EP measures (jitter/stagger) 7) Switching from Optronic to Active RADAR 8) Assess jamming 9) React to changes in EW environment	EDI Video demonstration Practical

Activity Steps	
	1) Instructor will explain conducting an investigate, ranging contacts and acquiring contacts to include: • Using Laser range finder vs RADAR • Passive RADAR, auto, optronic, manual optronic, passive RADAR and active RADAR with jammer When you would use, benefits of each and the environmental impact of each (20 min) 2) Instructor will demonstrate a walk through conducting an investigate, methods of ranging contacts and how to know you have acquired a contact. (20 min) 3) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (30 min)

	4) Students will then conduct investigate, range contacts and lock-up contacts using NCPM 1000 with instructor support. Instructor to circulate and watch each student conduct drills and provide feedback. (270 min) 5) Instructor will play a video to the class demonstrating how to switch from Optronic to Active RADAR. Instructor will explain how to recognize jamming and what reactions they must have (report) in the case of jamming. (10 min)

Resources	References
1) GO Operator 2) NCPM 1000 3) Optronic and Active RADAR video	1) NCPM 1000

Notes to Instructor

EO 001.02 - Execute gunnery drills and misfire procedures for 57mm

001.02.06 – 57mm Drills and Procedures

Conditions		Sequence
SEACOT		Following EO 001.02.05
Time	**Classification**	**Instructor**
1800	SECRET	RQ-S2 Instructor Require second instructor for 8+ students

Learning Objectives	Method
1) conduct 57mm Surface Secondary drills 2) conduct 57 mm surface primary drills 3) conduct 57mm air primary drills 4) conduct 57mm warnings shots drill 5) conduct misfire procedures (cold gun – primary locked on, secondary locked on, secondary fictitious, and failed misfire) 6) conduct communication reports	EDI Practical

Activity Steps	
	Misfire Drills (cold gun – primary locked on, secondary locked on, secondary fictitious, and failed misfire) 1) Instructor will explain 57mm Misfire drills. (30 min) 2) Instructor will demonstrate a walk through 57mm Misfire drills. (50 min) 3) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (110 min) 4) Students will then pair up and practice conducting 57mm Misfire drills using NCPM 1000 with instructor support. Instructor to circulate and watch each student conduct drills and provide feedback. (700 min) Surface Secondary Drills 5) Instructor will explain 57mm Surface Secondary drills. (10 min) 6) Instructor will demonstrate a walk through 57mm Surface Secondary drills. (20 min) 7) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (30 min) 8) Students will then pair up and practice conducting 57mm Surface Secondary drills using NCPM 1000 with instructor support. Instructor to circulate and watch each student conduct drills and provide feedback. (700 min) Surface Primary Drills 9) Instructor will explain 57mm Surface Primary drills. (10 min) 10) Instructor will demonstrate a walk through 57mm Surface Primary drills. (5 min)

11) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (5 min)
12) Students will then pair up and practice conducting 57mm Surface Primary drills using NCPM 1000 with instructor support. Instructor to circulate and watch each student conduct drills and provide feedback. (25 min)

Air Primary Drills

13) Instructor will explain 57mm Air primary drills. (10 min)
14) Instructor will demonstrate a walk through 57mm Air primary drills. (5 min)
15) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (5 min)
16) Students will then pair up and practice conducting 57mm Air primary drills using NCPM 1000 with instructor support. Instructor to circulate and watch each student conduct drills and provide feedback. (25 min)

Warning Shots Drills

17) Instructor will explain 57mm Warning shots drills. (10 min)
18) Instructor will demonstrate a walk through 57mm Warning shots drills. (10 min)
19) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (10 min)
20) Students will then pair up and practice conducting 57mm Warning shots drills using NCPM 1000 with instructor support. Instructor to circulate and watch each student conduct drills and provide feedback. (30 min)

Resources	References
1) NCPM 1000 2) GO Operator	1) NCPM 1000

Notes to Instructor

EO 001.02 - Execute gunnery drills and misfire procedures for 57mm

001.02.07 – ID Faults

Conditions		Sequence
SEACOT with briefing room		Following EO 001.02.06
Time	**Classification**	**Instructor**
50 min	SECRET	RQ-S2 Instructor

Learning Objectives	Method
1) Identify CEROS faults 2) understand fault identification colours (meatballs) 3) Process of checkload	lecture

Activity Steps	
	1) Instructor to conduct lecture on faults within CEROS, the various panels and colour codes. They are to use their system to show it and explain the impact and their reactions based on the various faults. (50 min).

Resources	References
1) Screenshot of faults (system overview)	1) CEROS operators manual

Notes to Instructor

EO 001.02 - Execute gunnery drills and misfire procedures for 57mm

001.02.08 – Naval Fire Support

Conditions		**Sequence**
Classified classroom		Following EO 001.02.07
Time	**Classification**	**Instructor**
50	Up to SECRET	RQ-S2 Instructor

Learning Objectives	**Method**
1) understand NGFS engagements 2) understand FCO roll in NGFS engagement	Guest lecture

Activity Steps	1) Guest lecture to present on Naval Fire Support engagements (50 min)

Resources	**References**
1) CSNI/SECRET workstation 2) Projector 3) PowerPoint lecture	1) NCPM 1000

Notes to Instructor

EO 001.02 - Execute gunnery drills and misfire procedures for 57mm

001.02.09 – Developmental

Conditions		**Sequence**
SEACOT		Following EO 001.02.08
Time	**Classification**	**Instructor**
850 min	SECRET	RQ-S2 Instructor

Learning Objectives	**Method**
1) Engage in live firing	Practical

Activity Steps	
	1) Students will be provided a CEROS console and support manning detailed in NWTP 2, where they must execute practice the duties of the FCO during gunnery drills and conduct misfire procedures. Remainder of the class to review the questions from the CEROS worksheet (from lesson 001.01A). 2) Each student will conduct a minimum of 2 x 35 min run, where each run will include: a. Deliberate PAC b. Surface Secondary engagement (hammerhead target) c. Primary Air 57mm (threat contact) d. Cold Gun Misfire Drills (secondary locked on, secondary fictitious, and failed misfire) 3) Instructor to provide feedback using EC 001.02 checklist for developmental purposes.

Resources	**References**
1) NCPM 1000	1) NCPM 1000

Notes to Instructor

EC 001.02 - Execute gunnery drills and misfire procedures for 57mm

Scope	Conditions	Sequence
EO 001.02	SEACOT	Following EO 001.02
Time	**Classification**	**Instructor**
500 min	SECRET	RQ-S2 Instructor

Assessment Steps	
	Input: Students will be provided a CEROS console and support manning detailed in NWTP 2, where they must execute the duties of the FCO during gunnery drills and conduct misfire procedures. Performance: students will conduct a 35 min run, where each run will include: • Deliberate PAC • Surface Secondary engagement (hammerhead target) • Primary Air 57mm (threat contact) • Cold Gun Misfire Drills (secondary locked on, secondary fictitious, and failed misfire) Output: Trainee must achieve 22/27 SATs on non-critical factors AND all critical factors to pass this EC. Feedback: Immediate verbal and written feedback provided using EC 001.02 checklist. 5 min allocated to reset the scenario and conduct debrief.

Resources	References
1) NCPM 1000	1) NCPM 1000

Notes to Instructor

Limitations	Nil
Impact of Failure	Retrain and reassess

Item	EC 001.02 EXECUTE GUNNERY DRILLS AND MISFIRE PROCEDURES FOR 57MM	SAT	UNSAT	Remarks
1. Conduct deliberate PAC				
Change HPI to 1.0		☐	☐	
Display setup IAW NCPM 1000 and MARTI 3221		☐	☐	
Verify allow transmit (RADAR)		☐	☐	
Log corrections IAW NCPM 1000 after each SALVO		☐	☐	
Conduct calculation of errors		☐	☐	
Input error of the day		☐	☐	
Verify error of the day IAW NCPM 1000		☐	☐	
Conduct system manipulation IAW NCPM 1000		☐	☐	
Conduct drill voice procedures IAW NCPM 1000		☐	☐	
2. Conduct Surface Secondary Engagement				
Conducts system manipulation IAW NCPM 1000		☐	☐	
Conducts voice procedures IAW NCPM 1000		☐	☐	
3. Conduct Secondary locked on misfire procedures IAW NCPM 1000				
Observe misfire indication		☐	☐	
Report misfire to FC Sup		☐	☐	
*Assigns gun to director		☐	☐	**CRITICAL**
Set SALVO size single		☐	☐	
Bring gun to red alert		☐	☐	
Prepare refire		☐	☐	
4. Conduct Primary Air 57 mm Engagement				
Report gun tracking on target to SWC IAW NCPM 1000		☐	☐	
Report hit/miss assessment to SWC IAW NCPM 1000		☐	☐	
5. Conduct secondary ficticious misfire procedures IAW NCPM 1000				
Observe misfire indication		☐	☐	
Report misfire to FC Sup		☐	☐	

Input lookout bearing into PAC window	☐	☐	
Set SALVO size single	☐	☐	
*Assigns gun to director	☐	☐	**CRITICAL**
Bring gun to red alert	☐	☐	
Prepare refire	☐	☐	
6. Conduct failed misfire procedures IAW NCPM 1000			
Unload upon FC Sup direction	☐	☐	
De-assign gun upon direction from SWC	☐	☐	
Turn RADAR to standby	☐	☐	

Score: /27 /2 CRITICAL
Trainee must achieve 22/27 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 001.03 - Execute missile drills and misfire procedures

001.03.01 – ESSM Details

Conditions		Sequence
Unclassified classroom		Following EC 001.02
Time	**Classification**	**Instructor**
50 min	unclassified	RQ-S2 Instructor

Learning Objectives	Method
1) ESSM capabilities and limitations a. Range/speed b. CWI 2) ESSM functionality a. HAW/THAW b. SASS mode 3) ESSM vulnerabilities	Lecture Video Case Study

Activity Steps	
	1) Instructor to lead a lecture on ESSM. They will discuss the caps/lims, functionality and vulnerabilities. (25 min) 2) Instructor will play a video of ESSM engagements. (5 min) 3) Instructor will provide a missile engagement report to the class. They will discuss what faults occurred, who's responsibility it was and the impact of such errors. (20 min)

Resources	References
1) PowerPoint lecture ESSM 2) DWAN computer 3) Projector 4) ESSM engagement video 5) Missile engagement report	1) CFCD 106

Notes to Instructor

EO 001.03 - Execute missile drills and misfire procedures

001.03.02 – ESSM system setup

Conditions		Sequence
SEACOT		Following EO 001.03.01
Time	**Classification**	**Instructor**
50 min	Up to SECRET	RQ-S2 Instructor

Learning Objectives	Method
1) ESSM buttonology 2) ESSM screen layout/configuration	EDI Practical

Activity Steps	
	1) Instructor will explain ESSM to include: • Buttonology • Assigning missiles • Screen layouts Explain what is contained in each page, when you are required to use each page and how to manipulate through the pages. (10 min) 2) Instructor will demonstrate a walkthrough of the buttonology and system setup. (15 min) 3) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (15 min) 4) Students will then have an opportunity to explore the system setup and ask instructor questions. (10 min)

Resources	References
1) NCPM 1000	1) NCPM 1000 2) CEROS user manual

Notes to Instructor

EO 001.03 - Execute missile drills and misfire procedures

001.03.03 – ESSM drills and procedures

Conditions		Sequence
SEACOT		Following EO 001.03,02
Time	**Classification**	**Instructor**
200 min	Up to SECRET	RQ-S2 Instructor

Learning Objectives	Method
1) conduct ESSM drill 2) conduct misfire procedures (DUD, CWI) 3) conduct communication reports	EDI Practical

Activity Steps

ESSM drill

1) Instructor will explain ESSM drills. (5 min)
2) Instructor will demonstrate a walk through ESSM drills. (10 min)
3) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (20 min)
4) Students will then pair up and practice conducting ESSM drills using NCPM 1000 with instructor support. Instructor to circulate and watch each student conduct drills and provide feedback. (145 min)

Misfire Drills (DUD & CWI)

5) Instructor will explain ESSM drills. (2 min)
6) Instructor will demonstrate a walk through of ESSM Misfire drills. (3 min)
7) Students will then be assigned a CEROS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (5 min)
8) Students will then pair up and practice conducting ESSM Misfire drills using NCPM 1000 with instructor support. Instructor to circulate and watch each student conduct drills and provide feedback. (10 min)

Resources	References
1) NCPM 1000	1) NCPM 1000

Notes to Instructor

When students pair up, one is to act as SWC while other practices drill

EC 001.03 - Execute missile drills and misfire procedures

Scope	Conditions	Sequence
EO 001.03	SEACOT	Following EO 001.03.03
Time	**Classification**	**Instructor**
200 min	SECRET	RQ-S2 Instructor

Assessment Steps	Input: Students will be provided a CEROS console and support manning detailed in NWTP 2, where they must execute the duties of the FCO during missile drills. Performance: students will conduct a 10 min run, where they will conduct ESSM Primary (threat contact) Output: The trainee must achieve 3/3 SATs to pass this EC. Feedback: Immediate verbal and written feedback provided using EC 001.03 checklist. 5 min allocated to reset the scenario and conduct debrief.

Resources	References
1) NCPM 1000	1) NCPM 1000

Notes to Instructor

Limitations	Nil
Impact of Failure	Retrain and reassess

Item	EO 001.03 EXECUTE MISSILE DRILLS AND MISFIRE PROCEDURES	SAT	UNSAT	Remarks
1. Conduct ESSM Primary Engagement				
Report missile tracking on target to SWC IAW NCPM 1000		☐	☐	
Reports hit/miss assessment for missile to SWC IAW NCPM 1000		☐	☐	
Check missile inventory for DUD (report as necessary)		☐	☐	

Score: /3

The trainee must achieve 3/3 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 001- Operate Combat Systems Equipment: CEROS

001 – Developmental

Conditions		Sequence
SEACOT		Following EC 001.03
Time	**Classification**	**Instructor**
1100 min	SECRET	RQ-S2 Instructor

Learning Objectives	Method
1) Operate CS equipment - CEROS	Practical

Activity Steps

1) Students will be provided opportunity to practice PC assessment under test conditions with feedback from the instructor. Students will be provided a CEROS console and support manning detailed in NWTP 2, where they must execute the duties of the FCO during gunnery drills, missile drills and misfire procedures with layered engagement.
2) Students will conduct a minimum of 2x 45 min run, where each run will include:
   - CUCA
   - Assume policy
   - Deliberate PAC
   - Surface Secondary engagement (hammerhead target)
   - ESSM Primary (threat contact)
   - Primary Air 57mm (threat contact)
   - Cold Gun Misfire Drills (secondary locked on, secondary fictitious, and failed misfire)
3) Instructor to provide feedback utilizing the PC 001 checklist.

Resources	References
1) NCPM 1000	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor

PC 001 - Operate Combat Systems Equipment: CEROS

Scope	Conditions	Sequence
PO 001	SEACOT	Following PO 001
Time	**Classification**	**Instructor**
600 min	SECRET	RQ-S2 Instructor and support manning

Scenario	Your ships is conducting 57mm firing exercises off the coast of Halifax/West Coast firing areas, when the operations team determines an air threat contact inbound. ESSM missed assessment forcing 57mm to engage in Primary.
Assessment Steps	Input: Students will be provided a CEROS console and support manning detailed in NWTP 2, where they must execute the duties of the FCO during gunnery drills, missile drills and misfire procedures with layered engagement. Performance: students will conduct a 45 min run, where each run will include: • CUCA • Assume policy • Deliberate PAC • Surface Secondary engagement (hammerhead target) • ESSM Primary (threat contact) • Primary Air 57mm (threat contact) • Cold Gun Misfire Drills (secondary locked on, secondary fictitious, and failed misfire) Output: Trainee must achieve 32/40 SATs on non-critical factors **AND** all critical factors to pass PC 001. Feedback: Immediate verbal and written feedback provided using PC 001 checklist. 5 min allocated to reset the scenario and conduct debrief.

Resources	References
1) NCPM 1000	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor
Ideal manning includes a SWC as well as GO operator

Limitations	
Impact of Failure	IAW NPTORD 4500-54

Item	PC 001 Operate Combat Systems Equipment: CEROS	SAT	UNSAT	Remarks
1. Conduct Closing Up Clearing away procedures				
Conduct Communications Check		☐	☐	
Conduct system setup IAW NCPM 1000		☐	☐	
Input environmentals and Nav data		☐	☐	
System Display setup IAW NCPM 1000		☐	☐	
Assign gun to FC channel		☐	☐	
Reports Closed up Cleared away to FC Sup		☐	☐	
2. Input Policy data to include:				
Select magazine		☐	☐	
Select forced recoil		☐	☐	
Set salvo size		☐	☐	
Assign missile		☐	☐	
3. Conduct deliberate PAC				
Change HPI to 1.0		☐	☐	
Display setup IAW NCPM 1000 and MARTI 3221		☐	☐	
Verify allow transmit (RADAR)		☐	☐	
Log corrections IAW NCPM 1000 after each SALVO		☐	☐	
Conduct calculation of errors		☐	☐	
Input error of the day		☐	☐	
Verify error of the day IAW NCPM 1000		☐	☐	
Conduct system manipulation IAW NCPM 1000		☐	☐	
Conduct drill voice procedures IAW NCPM 1000		☐	☐	
4. Conduct Surface Secondary Engagement				

Conducts system manipulation IAW NCPM 1000	☐	☐	
Conducts voice procedures IAW NCPM 1000	☐	☐	
5. Conduct Secondary locked on misfire procedures IAW NCPM 1000			
Observe misfire indication	☐	☐	
Report misfire to FC Sup	☐	☐	
*Assigns gun to director	☐	☐	**CRITICAL**
Set SALVO size single	☐	☐	
Bring gun to red alert	☐	☐	
Prepare refire	☐	☐	
6. Conduct ESSM Primary Engagement			
Report missile tracking on target to SWC IAW NCPM 1000	☐	☐	
Reports hit/miss assessment for missile to SWC IAW NCPM 1000	☐	☐	
Check missile inventory for DUD (report as necessary)	☐	☐	
7. Conduct Primary Air 57 mm Engagement			
Report gun tracking on target to SWC IAW NCPM 1000	☐	☐	
Report hit/miss assessment to SWC IAW NCPM 1000	☐	☐	
8. Conduct secondary fictitious misfire procedures IAW NCPM 1000			
Observe misfire indication	☐	☐	
Report misfire to FC Sup	☐	☐	
Input lookout bearing into PAC window	☐	☐	
Set SALVO size single	☐	☐	
*Assigns gun to director	☐	☐	**CRITICAL**
Bring gun to red alert	☐	☐	
Prepare refire	☐	☐	
9. Conduct failed misfire procedures IAW NCPM 1000			
Unload upon FC Sup direction	☐	☐	

De-assign gun upon direction from SWC	☐	☐	
Turn RADAR to standby	☐	☐	

Score: /40 /2 CRITICAL

Trainee must achieve 32/40 SATs on non-critical factors **AND** all critical factors to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 – Operate Combat Systems Equipment: CIWS

1. **Performance**: Operate Combat Systems Equipment: CIWS (B00080, D00060, D00010)

2. **Conditions**:

   a. Given:

      (1) Resources;
      CPF layout handout
      CIWS Handout

      (2) References;
      NCPM 1000
      NAVORD 4995-0
      (3) Supervision.

   b. Denied:

      (1) Nil

3. **Environment**:

      (1) At Sea/Alongside Day or Night
      (2) Trainer/Simulator Day or Night

4. **Standard**: The Fire Control Operator shall operate combat systems equipment: CIWS (B00080, D00060, D00010) by:

   a. Executing preparation for CIWS gunnery drills by:

      (1) Conducting closing up, clearing away procedures IAW NCPM 1000;
      (2) Assuming policy to include:
         (a) Adhering to EMCON restrictions (B00120); and
         (b) Setting parameters set by the SWC;
      (3) Posting ships weapons firing out of bounds signage; and
      (4) Preparing surface targets for gunnery exercises (D00020);

   b. Executing gunnery drills and misfire procedures for CIWS by:

      (1) Operating CIWS for gunnery drills by:
         (a) Conducting system tracking assessments;
         (b) Communicating reports;
         (c) Performing threat analysis (C00540);
         (d) Identifying faults to include:
            i. Bore Loaded; and
            ii. General faults; and
         (e) Engage in live firing.

5. **Training Limitations**:

EO 002.01 - Execute preparation for CIWS gunnery drills

002.01.01 – CIWS Theory

Conditions		Sequence
Unclassified classroom		Following PC 001
Time	**Classification**	**Instructor**
150 min	unclassified	RQ-S2 Instructor

Learning Objectives	Method
1) Introduction to CIWS 2) CIWS capabilities and limitations a. Ranges b. Rate of fire c. No-engage sectors d. Radar capabilities e. Tracking capabilities f. Tracking modes between air and surface modes 3) CIWS functionality	Lecture Team Quiz Student Reading Diagram Labeling

Activity Steps	
	1) Students are asked to each create 3 questions during the lesson, must include answers. Instructor to conduct a lecture on CIWS, caps/lims, functionality and the alarm. Lesson to include a sound clip of CIWS alarm and a video of CIWS firing sequence. (70 min) 2) Students submit their questions to the instructor and split into two teams. 3) Instructor to then conduct a team quiz activity, tracking points and using the CIWS alarm sound clip to indicate any incorrect answers. (30 min) 4) Students to review NAVORD 4995-0 section 3 (Internal safety) (25 min) 5) Students are then provided a print out of a CPF plan. Students are to indicate where OOB signs for various weapons firing serials go on a plan of a CPF AND create a list of the spaces. (25 min) ***** reminder to students to study the OOB signage locations for EC 002.01**

Resources	References
1) CIWS lecture 2) CIWS alarm sound clip 3) CIWS firing sequence video 4) NAVORD 4995-0 5) CPF layout handout	1) NAVORD 4995-0 2) CIWS user manual 3) NCPM 1000

EO 002.01 - Execute preparation for CIWS gunnery drills

002.01.02 – CIWS Preparations

Conditions		Sequence
Trainer/ship (Primary) SEACOT (secondary)		Following EO 002.01.01
Time	**Classification**	**Instructor**
800 min	Up to SECRET	RQ-S2 Instructor

Learning Objectives	Method
1) CIWS buttonology 2) CIWS screen layout/configuration 3) Conducting Closing Up and Clearing Away 4) Assuming Policy a. Setting SWC parameters b. System manipulation 5) Inputting Policy orders (burst length, ARS, no-engage, first-hit, search sectors)	EDI Practical

Activity Steps	
	1) Students broken into two groups (if class size 8+). ½ class to conduct lesson at Trainer/ship, while other ½ is at SEACOT (or admin day). 2) Trainer/ship Instructor to conduct a lecture on: functionality of mount, feed system, radar, rounds, FIK location, sector hold back tool, air tank, cut-outs (50 min) 3) RQ-S2 Instructor will then conduct a lesson to explain and demonstrate CIWS buttonology, screen layout, how to conduct CUCA and Assuming Policy. (50 min) 4) Students are to each 1 by 1 have an opportunity to be on the console with instructor direction to conduct system manipulation, practice CUCA and assuming policy. Remainder of the students are to work on a CIWS worksheet containing: button locations and CIWS theory questions. (300 min) For MAX load, total time is 800 min (as 1 day admin 1 day Trainer/ship) due to equipment limitations.

Resources	References
1) NCPM 1000 2) Transport to and from site visit 3) Trainer/ship (armourment technician)	1) NCPM 1000 2) CIWS manual

Notes to Instructor
If unable to get a second instructor to take ½ class to SEACOT, ½ class will take the weeks worth of BAT/TAT and PT periods as their Admin day, and attend Trainer/ship the following day

If a software update to SEACOT is complete, lesson can be 100% conducted at SEACOT vice Trainer/ship.

EC 002.01 - Execute preparation for CIWS gunnery drills

Scope	Conditions	Sequence
EO 002.01	Trainer/ship	Following EO 002.01
Time	**Classification**	**Instructor**
200 min	unclassified	RQ-S2 Instructor

Assessment Steps	
	Input: Students will be provided a CIWS console, where they must execute the duties of the FCO in preparation for CIWS firing exercises. Performance: students will conduct a 15 min run, where each run will include: • CUCA • Assume policy • Verbally indicate OOB signage locations for CIWS firing Output: The trainee must achieve 11/13 SATs to pass this EC. Feedback: Immediate verbal and written feedback provided using EC 002.01 checklist.

Resources	References
1) NCPM 1000	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor

Limitations	Nil
Impact of Failure	Retrain and reassess

Item	EC 002.01 Execute preparation for CIWS gunnery drills	SAT	UNSAT	Remarks
1. Conduct Closing Up Clearing away procedures				
Conduct Communications Check		☐	☐	
Confirm RCP key inserted		☐	☐	
Conduct system setup IAW NCPM 1000				
- Select EO standby		☐	☐	
- Select EO ready		☐	☐	
- Select search and train & elevate the mount		☐	☐	
- Revert to EO standby		☐	☐	
Reports Closed up Cleared away to FC Sup		☐	☐	
2. Input Policy data IAW AWWD to include:				
Set ARS with code 2		☐	☐	
Input search sector data		☐	☐	
Input no engage sector data		☐	☐	
Input min/max range rate		☐	☐	
Report standing to		☐	☐	
3. Verbally indicate OOB signage locations for weapons firing				
Verbally indicate **ALL** CIWS signage locations ☐ Ladders to Hanger Mezzanines x 2 ☐ Hangar lobby doors x 2 ☐ LSO CMPT hatch ☐ AX hatch		☐	☐	

Score: **/ 13** The trainee must achieve 11/13 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.02 - Execute gunnery drills and misfire procedures for CIWS

002.02.01 – CIWS drills and procedures

Conditions		Sequence
SEACOT		Following EC 002.01
Time	**Classification**	**Instructor**
150 min	Up to SECRET	RQ-S2 Instructor

Learning Objectives	Method
1) Aquire target utilizing CIWS 2) Conduct CIWS tracking 3) Conduct communication reports 4) conduct CIWS surface gunnery drills 5) execute live firing 6) identify faults and responses (bore loaded, general, tx error)	EDI Practical

Activity Steps	
	1) Instructor will explain CIWS target acquisition, tracking and drills to include: • Aquire target using CIWS • Target tracking using CIWS • CIWS drills and procedures • Responses to faults (bore loaded, general faults and tx error) (15 min) 2) Instructor will demonstrate a walk through of CIWS drills and procedures. (30 min) 3) Students will then be assigned a CIWS console, where they will follow along on their own systems while the instructor does a guided walk through demonstrating again on the projected instructor console. (50 min) 4) Students will then pair up and practice CIWS drills and procedures, with instructor providing feedback throughout. (55 min)

Resources	References
1) NCPM 1000 2) CUD	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor

EC 002.02 - Execute gunnery drills and misfire procedures for CIWS

Scope	Conditions	Sequence
EO 002.02	SEACOT	Following EO 002.02
Time	**Classification**	**Instructor**
200 min	unclassified	RQ-S2 Instructor

Assessment Steps	

	Input: Students will be provided a CIWS console, where they must execute the duties of the FCO during CIWS firing exercises.
	
	Performance: students will conduct a 15 min run, where each run will include:
	• Surface engagement (hammerhead target)
	  ○ Apply hit corrections
	  ○ Alter SALVO size
	  ○ RF ranging
	  ○ Passive ranging
	  ○ Manual ranging
	  ○ FC bypass
	
	Output: The trainee must achieve 16/20 SATs to pass this EC.
	
	Feedback: Immediate verbal and written feedback provided using EC 002.02 checklist.

Resources	References
1) NCPM 1000	1) NCPM 1000

Notes to Instructor

Limitations	Nil
Impact of Failure	Retrain and reassess

Item	EC 002.02 Execute gunnery drills and misfire procedures for CIWS	SAT	UNSAT	Remarks
1. Conduct surface engagement (hammerhead target)				
Slew mount to target (manually or BRG enter)		☐	☐	
Press trigger to acquire target		☐	☐	
Acquire lock-on by:				
• Select SURF RF ON (Uses RF +/- as required)		☐	☐	
• Selects Passive Ranging		☐	☐	
• Selects Manual Ranging		☐	☐	
• Selects FC Bypass		☐	☐	
Select ENGAGE ON		☐	☐	
Report target IAW NCPM 1000		☐	☐	
Upon receiving engage order, press FIRE button and Reports CIWS shoot IAW NCPM 1000		☐	☐	
Reports hit/miss assessment		☐	☐	
Adjust aim (apply corrections)		☐	☐	
Report CIWS corrected		☐	☐	
press FIRE button and Reports CIWS shoot IAW NCPM 1000		☐	☐	
Reports hit/miss assessment		☐	☐	
Input SALVO size		☐	☐	
press FIRE button and Reports CIWS shoot IAW NCPM 1000 and policy		☐	☐	
Upon receipt of “CEASE FIRE, MAKE CIWS SAFE”, stop shooting		☐	☐	
Upon receipt of “BREAK ENGAGE”, select BREAK ENGAGE		☐	☐	

Revert back to ordered control mode (policy)	☐	☐	
Report selected control mode	☐	☐	

Score: **/ 20** The trainee must achieve 16/20 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 - Operate Combat Systems Equipment: CIWS

002- Development

Conditions		Sequence
SEACOT		Following EC 002.02
Time	**Classification**	**Instructor**
600 min	Up to SECRET	RQ-S2 Instructor

Learning Objectives	Method
1) engage in live CIWS firing	Practical

Activity Steps

1) Students will be provided opportunity to practice PC assessment under test conditions with feedback from the instructor. Students will be provided a CIWS console and support manning detailed in NWTP 2, where they must execute the duties of the FCO during CIWS engagement.
2) students will conduct a minimum of 2 x 25 min run, where each run will include:
   - CUCA
   - Assume policy
   - Surface engagement (hammerhead target)
     - Apply hit corrections
     - Alter SALVO size
     - RF ranging
     - Passive ranging
     - Manual ranging
     - FC bypass
3) Instructor to provide feedback utilizing the PC 002 checklist.

Resources	References
1) NCPM 1000	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor

Instructor will explain (and reset the system on behalf of the student) IOT reduce bad habits from trainer.

PC 002 – Operate Combat Systems Equipment: CIWS

Scope	Conditions	Sequence
PO 002	SEACOT	Following PO 002
Time	**Classification**	**Instructor**
300 min	unclassified	RQ-S2 Instructor

Scenario	Your ships is conducting CIWS firing exercises off the coast of Halifax/West Coast firing areas.

Assessment Steps	Input: Students will be provided a CIWS console, where they must execute the duties of the FCO CIWS firing exercises.
	
	Performance: students will conduct a 25 min run, where each run will include:
	• CUCA
	• Assume policy
	• Surface engagement (hammerhead target)
	  ○ Apply hit corrections
	  ○ Alter SALVO size
	  ○ RF ranging
	  ○ Passive ranging
	  ○ Manual ranging
	  ○ FC bypass
	
	Output: The trainee must achieve 26/32 SATs to pass this PC.
	
	Feedback: Immediate verbal and written feedback provided using PC 002 checklist.

Resources	References
1) NCPM 1000	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor

Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54

Item	PC 002 – OPERATE COMBAT SYSTEMS EQUIPMENT: CIWS	SAT	UNSAT	Remarks
1. Conduct Closing Up Clearing away procedures				
Conduct Communications Check		☐	☐	
Confirm RCP key inserted		☐	☐	
Conduct system setup IAW NCPM 1000				
• Select EO standby		☐	☐	
• Select EO ready		☐	☐	
• Select search and train & elevate the mount		☐	☐	
• Revert to EO standby		☐	☐	
Reports Closed up Cleared away to FC Sup		☐	☐	
2. Input Policy data IAW AWWD to include:				
Set ARS with code 2		☐	☐	
Input search sector data		☐	☐	
Input no engage sector data		☐	☐	
Input min/max range rate		☐	☐	
Report standing to		☐	☐	
3. Conduct surface engagement (hammerhead target)				
Slew mount to target (manually or BRG enter)		☐	☐	
Press trigger to acquire target		☐	☐	
Acquire lock-on by:				
• Select SURF RF ON • (Uses RF +/- as required)		☐	☐	
• Selects Passive Ranging		☐	☐	
• Selects Manual Ranging		☐	☐	
• Selects FC Bypass		☐	☐	
Select ENGAGE ON		☐	☐	
Report target IAW NCPM 1000		☐	☐	
Upon receiving engage order, press FIRE button and Reports CIWS shoot IAW NCPM 1000		☐	☐	

Reports hit/miss assessment	☐	☐	
Adjust aim (apply corrections)	☐	☐	
Report CIWS corrected	☐	☐	
press FIRE button and Reports CIWS shoot IAW NCPM 1000	☐	☐	
Reports hit/miss assessment	☐	☐	
Input SALVO size	☐	☐	
press FIRE button and Reports CIWS shoot IAW NCPM 1000 and policy	☐	☐	
Upon receipt of "CEASE FIRE, MAKE CIWS SAFE", stop shooting	☐	☐	
Upon receipt of "BREAK ENGAGE", select BREAK ENGAGE	☐	☐	
Revert back to ordered control mode (policy)	☐	☐	
Report selected control mode	☐	☐	

Score: **/ 32**

The trainee must achieve 26/32 SATs to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

CHAPTER 3: ASSESSMENT OF TRAINEE

GENERAL

1. All Performance Checks (PCs) are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of PRACTICAL or WRITTEN assessment and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Each EO must have at least one EC. EC results do not factor into a trainee's successful completion of the PO.

3. Performance Checks (PCs). PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be completed IAW the standards delineated in this QSP and IAW CFITES Volume 7 (Evaluation).

PROGRESS MONITORING

5. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Volume 6 Conduct of Instruction Programmes.

6. Continuous monitoring of trainee progress is required to provide the following:

    a. Early warning of trainee difficulties;
    b. Feedback on the effectiveness of training;
    c. Information for the Training Review Board (TRB) (if required); and
    d. Information to assist in the completion of the CF 377 Course Report.

7. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment (TE) SOs or SOPs.

8. PC failures and supplemental PC failures are to be handled IAW TE SOs and SOPs. Supplemental ECs and PCs shall not be given no less than 24 hours of the trainee being informed of their EC or PC failure.

TRAINING REVIEW BOARD

9. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54 Training Review Board Process.

COURSE REPORTS

10. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51.

11. Trainees will be graded pass or fail. NPTORD 4500-51 Course Reporting and Certificates, provides direction and guidelines for Letter Grading and Ranking.

12. Narratives will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and/or incomplete training must also be included in the course report. A record of trainee counselling is to be maintained by the TE.

Task List

Task #	Task Statement	Trg / NT	PO
A00010	Authenticate communication transmissions	NT	-
A00020	Perform communication checks	NT	-
A00030	Configure communication terminal for operation	NT	-
A00040	Extract information from Communications Plans (COMPLANS), Emission Control (EMCON) Plans and Frequency Plans	NT	-
A00060	Communicate using tactical voice procedures	NT	-
B00050	Perform evidence gathering	NT	-
B00070	Perform Intelligence Gathering	NT	-
B00080	Operate combat systems (CS) equipment	T	001/002
B00090	Evaluate intelligence information	NT	-
B00110	Maintain stateboards	NT	-
B00120	Adhere to EMCON restrictions	T	001/002
B00140	Adhere to Emissions Control (EMCON) restrictions	NT	-
B00160	Operate Intelligence gathering Equipment	NT	-
B00170	Perform pre-sail checks on equipment	NT	-
B00190	Prepare EMCON Table	NT	-
C00020	Conduct Electronic Attacks	NT	-
C00050	Assess jamming	T	001
C00070	Carryout Electronic Protect (EP) measures	T	001
C00480	Perform Electronic Attack (EA) decoys loading procedures	NT	-
C00540	Perform threat analysis	T	001/002
C00580	React to changes in the EW environment	T	001
D00010	Operate Anti Surface Warfare (ASuW) weapons and sensors	T	001/002
D00020	Prepare surface targets for gunnery exercises	T	001/002
D00060	Operate Anti Air Warfare (AAW) weapons and sensors	T	001/002
D00080	Operate FC systems and associated equipment	T	001
F00080	Conduct/Participate in technical evaluations, tests and trials	NT	-
F00180	Identify Maritime, Air, and Land based platforms	NT	-

Record of Proceedings

Dates: 27 – 31 May 2019

Location: CFB Stadacona, Building S15 rm 214

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
McPhee	S.	PO2	00115	NTDC(A)
Parsons	S.	PO2	00115	NTDC(A)
Moffat	T.	MS	00115	NFS(A)
MacArthur	D.	MS	00115	NTDC(A)
Carr	D.	MS	00115	NTDC(A)
Johnson	T.D.	S1	00115	MON
Drain	S.O.	S1	00115	CHA
Jawornicki	B.	SLt	00211	NTDC(A)

Working Group Objective:

The objective of this working group is to conduct updates to the RQ-S2 NESOP QSP. Issues were noted when the Combat Training Review was conducted Jan-Feb 2019 where the recommendation to conduct a QSP review board was recommended. As a result, the QSP was updated to reflect the changes identified and input to the FireFly format.

Specification Task

No Train Decisions:

Task #	Task Statement	Substantiation
A00010	Authenticate communication transmissions	This task is trained at a lower rank level
A00020	Perform communication checks	This task is trained at a lower rank level
A00030	Configure communication terminal for operation	This task is trained at a lower rank level
A00040	Extract information from Communications Plans (COMPLANS), Emission Control (EMCON) Plans and Frequency Plans	This task is trained at a lower rank level
A00060	Communicate using tactical voice procedures	This task is trained at a lower rank level
B00050	Perform evidence gathering	Board decision was to remove this task as NESOP's are not part of the evidence gathering team.
B00070	Perform Intelligence Gathering	Board decision was to remove this task as NESOP's are not part of the intelligence gathering team.

B00090	Evaluate intelligence information	Board decision was to remove this task as NESOP's are not part of the intelligence gathering team.
B00110	Maintain stateboards	This task is trained at a lower rank level
B00140	Adhere to Emissions Control (EMCON) restrictions	This task is trained at a lower rank level
B00160	Operate Intelligence gathering Equipment	Board decision was to remove this task as NESOP's are not part of the intelligence gathering team.
B00170	Perform pre-sail checks on equipment	This task is trained at a lower rank level
B00190	Prepare EMCON Table	This task is trained at a lower rank level
C00020	Conduct Electronic Attacks	This task is trained at a higher rank level
C00480	Perform Electronic Attack (EA) decoys loading procedures	This task is trained at a lower rank level
F00080	Conduct/Participate in technical evaluations, tests and trials	This task is trained at a higher rank level
F00180	Identify Maritime, Air, and Land based platforms	This task is trained at a lower rank level

Discussions / Decisions:

1. Secondary procedures is being input to NCPM 1000 and therefore was required to be instructed to the RQ-S2 NESOPs. NTDC(A) is adding the procedures to the reference.
2. Laser Range Finder and IR are new features added to the RADAR (director) and therefore as new capability is an addition to the training requirements. This has been added to EO 001.02.

Course History:

Costing Information

Include information about cost in whichever format is applicable (**Not to be deleted**).

Course: RQ-NESOP-S2	Serial:					
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	4	$1,600.00		$6,400	$6,400	$64,000.00
R&Q	4	$78.75	40	$12,600	$12,600	$126,000.00
Incidentals	4	$17.30	40	$2,768	$2,768	$27,680.00
Subtotal				**$21,768.00**	**$21,768.00**	**$217,680.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	10	$50.00	1	$500	$500	$5,000.00
Classroom (square meters)	30	$13.00	40	…	…	…
Computer requirements	12	$0.00	1	$0	$0	$0.00
IT Equip (Proj, Smart board etc)	3	$0.00	1	$0	$0	$0.00
Instructional Materials (Printed / Handouts etc)	8	$30.00	1	$240	$240	$2,400.00
Subtotal				**$740**	**$740**	**$7,400.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1) (incremental)5	0	$0.00	0	$0	$0	$0.00
Instructor (Lt(N)) (incremental)5	0	$0.00	0	$0	$0	$0.00
Incremental Civilian Instructor	0	0	0	$0	$0	$0.00
		subtotal		**$0.00**	**$0.00**	**$0.00**
	Total Estimated Costs			**$22,508**	**$22,508**	**$225,080**

Assessment Plan

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess student's competence in executing preparation for 57mm gunnery drills.	Practical	Student must obtain 10 / 12 SATs IAW EC 001.01 Checklist
EC 001.02	EO 001.02	To assess student's competence in executing gunnery drills and misfire procedures for 57mm	Practical	Student must obtain 22 / 27 SATs IAW EC 001.02 Checklist
EC 001.03	EO 001.03	To assess student's competence in executing missile drills and misfire procedures.	Practical	Student must obtain 3 /3 SATs IAW EC 001.3 Checklist
PC 001	PO 001	To assess student's competence in operating combat systems equipment: CEROS	Practical	Student must obtain 32 /40 SATs IAW PC 001 Checklist
EC 002.01	EO 002.01	To assess student's competence in executing preparation for CIWS gunnery drills	Practical	Student must obtain 11 /13 SATs IAW EC 002.01 Checklist
EC 002.02	EO 002.02	To assess student's competence in executing gunnery drills and misfire procedures for CIWS	Practical	Student must obtain 16 /20 SATs IAW EC 002.02 Checklist
PC 002	PO 002	To assess student's competence in operating combat systems equipment: CIWS	Practical	Student must obtain 26 /32 SATs IAW PC 002 Checklist

TEL Development Opportunities

Title	Type	Description of Requirement	Lesson	SOR #
ESSM engagement	Video	Allows students to witness different modes of ESSM engagements HAW/THAW/SASS	001.03A	
CIWS theory	Video	Allows students to witness different operation of CIWS that includes firing and alarms	002.01A	

Scheduling Information

Scheduling Annex (Firefly) v1.xlsx

Scheduling Annex - COURSE NAME													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time	Location	DWAN	CSNI	SSA	Proj	TEL	Sec Class	WB	Desks	Type	Qual	Sec Cl
PO 001 - Operate combat systems equipment: CEROS													
001.01.01	200	Classroom	1	0	0	1		UNC	1	12	Mil	RQ-S2 NESOP	Sec
001.01.02	50	Classroom	0	0	0	0		UNC	1	12	Mil	RQ-S2 NESOP	Sec
001.01.03	50	Classroom	0	0	0	0		UNC	1	12	Mil	RQ-S2 NESOP	Sec
001.01.04	50	Classroom	0	0	0	0		UNC	1	12	Mil	RQ-S2 NESOP	Sec
001.01.05	200	Target Cell	0	0	0	0		UNC	0	0	Mil		
001.01.06	150	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
001.01.07	200	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
EC 001.01	250	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
001.02.01	100	Classroom	13	0	0	0		UNC	1	12	Mil	RQ-S2 NESOP	Sec
001.02.02	50	Classroom	13	0	0	0		UNC	1	12	Mil	RQ-S2 NESOP	Sec
001.02.03	150	Trainer/ship	0	0	0	0		UNC	0	0	Mil		
001.02.04	700	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
001.02.05	350	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
001.02.06	1800	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
001.02.07	50	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
001.02.08	50	Classroom	0	1	1	1		SEC	0	13	Mil		Sec
001.02.09	850	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec

EC 001.02	500	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
001.03.01	50	Classroom	13	0	0	1		UNC	1	13			
001.03.02	50	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
001.03.03	200	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
EC 001.03	200	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
P0 001	1100	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
PC 001	600	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
PO 002 - Operate combat systems equipment: CIWS													
002.01.01	150	Classroom	0	0	0	0		UNC	0	12	Mil	RQ-S2 NESOP	Sec
002.01.02	800	Trainer/ship /SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
EC 002.01	200	Trainer/ship	0	0	0	0		UNC	0	12	Mil	RQ-S2 NESOP	Sec
002.02.01	150	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
EC 002.02	200	SEACOT	0	0	0	0		UNC	0	12	Mil	RQ-S2 NESOP	Sec
PO 002	600	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec
PC 002	300	SEACOT	0	0	0	0		SEC	0	12	Mil	RQ-S2 NESOP	Sec

Recommended Schedule

Time	Day 1	Day 2	Day3	Day 4	Day 5
Period 1 0800-0850	Expectations /Admin ISSO ,IT, Command Intro	001.01.01 Intro to CEROS	001.01.06 CEROS Manipulation	EC 001.01 Execute Preps for 57mm	001.02.02 57mm Caps and Lims /Ammo types
Period 2 0850-0940		001.01.02 Readiness and Preparedness			001.02.03 57mm Famil tour
0940-0950	**Break**				
Period 3 0950-1040	**BAT**	001.01.03 EMCON/RADHAZ	001.01.06 CEROS Manipulation	EC 001.01 Execute Preps for 57mm	001.02.03 57mm Famil tour
Period 4 1040-1130		001.01.04 Out of Bounds	**PT**		
1130-1230	**Lunch**				
Period 5 1230-1320	**TAT**	001.01.05 Prepare Surface targets	001.01.07 CUCA/Policy	EC 001.01 Execute Preps for 57mm	**PT**
Period 6 1320-1410				**PT**	001.02.04 PAC Practical
1410-1420	**Break**				
Period 7 1420-1510	001.01.01 Intro to CEROS	001.01.05 Prepare Surface targets	001.01.07 CUCA/Policy	001.02.01 57mm PAC Theroy	001.02.04 PAC Practical
Period 8 1510-1600	001.01.01 Intro to CEROS				

Recommended Schedule

Time	Day 6	Day 7	Day 8	Day 9	Day 10
Period 1 0800-0850	001.02.04 PAC Practical	001.02.04 PAC Practical	001.02.05 Investigate and Acquire	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures
Period 2 0850-0940					
0940-0950	**Break**				
Period 3 0950-1040	001.02.04 PAC Practical	001.02.04 PAC Practical	001.02.05 Investigate and Acquire	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures
Period 4 1040-1130				**PT**	
1130-1230	**Lunch**				
Period 5 1230-1320	001.02.04 PAC Practical	PT	PT	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures
Period 6 1320-1410		001.02.05 Investigate and Acquire	001.02.06 57mm Drill and procedures		
1410-1420	**Break**				
Period 7 1420-1510	001.02.04 PAC Practical	001.02.05 Investigate and Acquire	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures
Period 8 1510-1600	BAT				TAT

Recommended Schedule

Time	Day 11	Day 12	Day 13	Day 14	Day 15
Period 1 0800-0850	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures	001.02.09 Developmental	001.02.09 Developmental
Period 2 0850-0940					
0940-0950	**Break**				
Period 3 0950-1040	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures	001.02.09 Developmental	001.02.09 Developmental
Period 4 1040-1130	**BAT**	**PT**	**PT**	**PT**	**TAT**
1130-1230	**Lunch**				
Period 5 1230-1320	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures	001.02.09 Developmental	001.02.09 Developmental
Period 6 1320-1410					
1410-1420	**Break**				
Period 7 1420-1510	001.02.06 57mm Drill and procedures	001.02.06 57mm Drill and procedures	001.02.07 ID Faults	001.02.09 Developmental	001.02.09 Developmental
Period 8 1510-1600			001.02.08 NFS		

Recommended Schedule

Time	Day 16	Day 17	Day 18	Day 19	Day 20
Period 1 0800-0850	001.02.09 Developmental	EC 001.02 Gunnery Drills/Misfire Procedures	TAT	EC 001.03 Missile Drills /Misfire Procedures	001 Developmental
Period 2 0850-0940			001.03.01 ESSM Details		
0940-0950	**Break**				
Period 3 0950-1040	001.02.09 Developmental	EC 001.02 Gunnery Drills/Misfire Procedures	001.03.02 ESSM setup	EC 001.03 Missile Drills /Misfire Procedures	001 Developmental
Period 4 1040-1130	**PT**		**PT**	001 Developmental	
1130-1230	**Lunch**				
Period 5 1230-1320	EC 001.02 Gunnery Drills/Misfire Procedures	EC 001.02 Gunnery Drills/Misfire Procedures	001.03.03 ESSM Drills/Procedures	001 Developmental	001 Developmental
Period 6 1320-1410					
1410-1420	**Break**				
Period 7 1420-1510	EC 001.02 Gunnery Drills/Misfire Procedures	BAT	001.03.03 ESSM Drills/Procedures	001 Developmental	001 Developmental
Period 8 1510-1600		PT			

Recommended Schedule

Time	Day 21	Day 22	Day 23	Day 24	Day 25
Period 1 0800-0850	001 Developmental	001 Developmental	002.01.01 CIWS Theory	002.01.02 CIWS Preps	002.01.02 CIWS Preps
Period 2 0850-0940					
0940-0950	**Break**				
Period 3 0950-1040	001 Developmental	PC 001	002.01.01 CIWS Theory	002.01.02 CIWS Preps	002.01.02 CIWS Preps
Period 4 1040-1130	**BAT**		**PT**		**TAT**
1130-1230	**Lunch**				
Period 5 1230-1320	001 Developmental	PC 001	002.01.02 CIWS Preps	PT	PT
Period 6 1320-1410				002.01.02 CIWS Preps	002.01.02 CIWS Preps
1410-1420	**Break**				
Period 7 1420-1510	001 Developmental	PC 001	002.01.02 CIWS Preps	002.01.02 CIWS Preps	002.01.02 CIWS Preps
Period 8 1510-1600					EC 002.01

Recommended Schedule

Time	Day 26	Day 27	Day 28	Day 29	Day 30
Period 1 0800-0850	EC 002.01	EC 002.02	002 Developmental	PC 002	TAT GRAD
Period 2 0850-0940					
0940-0950	**Break**				
Period 3 0950-1040	EC 002. 01	EC 002.02	002 Developmental	PC 002	BAT PCD
Period 4 1040-1130	**BAT**				
1130-1230	**Lunch**				
Period 5 1230-1320	002.02.01 CIWS Drills/Procedures	002 Developmental	002 Developmental	PC 002	
Period 6 1320-1410					
1410-1420	**Break**				
Period 7 1420-1510	002.02.01 CIWS Drills/Procedures	002 Developmental	002 Developmental	PT	
Period 8 1510-1600	PT				

Plan, Train, Do

PO	EO	Maintain Readiness	Plan	Train	Execute
001	01			Execute preparation for 57mm gunnery drills	
	02			Execute gunnery drills and misfire procedures for 57mm	
	03			Execute missile drills and misfire procedures	
002	01			Execute preparation for CIWS gunnery drills	
	02			Execute gunnery drills and misfire procedures for CIWS	

References

Ref Code	Publication
1	57mm Mk 3 Handbook
2	CEROS operators manual
3	CFCD 106
4	CFCD 106 MARTI 3221
5	http://targetsystems.qinetiq.com/static/media/files/Hammerhead_DS-HH-V01-A_4.pdf
6	http://targetsystems.qinetiq.com/static/media/files/HSITT_Mk_II.pdf
7	http://www.geodatasys.com/TECH/TGT/Tomato.pdf
8	https://phet.colorado.edu/sims/html/projectile-motion/latest/projectile-motion_en.html
9	NAVORD 4995-0
10	NCPM 1000
11	Sea trg group (atlantic) webpage for Rapid Survey Zones
12	SEMS (Safety and Enviroment Mangament systems)
13	SSO's